

Amendment No. 3 (09/2026))
To
AIS 039 (Rev 1): Electric Power Train Vehicles– Measurement of Electrical
Energy Consumption

1.0 Scope

Add following note below scope of the standard :

Note : With effective date of applicability of AIS-175 for ICE vehicles, Electrical energy consumption expressed in Wh/km for Electric Power Train Vehicles of category M1 & M2 with GVW up to 3.5 T shall be measured as per the procedure laid down in AIS-175 and for categories L, M2 with GVW above 3.5 T, M3 & N shall be as per AIS-039 (Rev.1). However, for CAFÉ calculation purpose, Additional Electrical energy consumption test may be conducted in MIDC as per existing procedure based on OEM request till the time CAFÉ migrates from MIDC to WLTP

2.0 Clause No 3.4 (b)

Substitute following text for first paragraph existing text:

(b) Application of number of World-harmonised Motorcycle Test Cycle (WMTC) stage 2 cycles for L1 and L2 category vehicles as per AIS 137 Part 1 and details as per below table.

Sr. No	Category of Vehicle	Max Speed	Test Cycle	Application of number of Cycles for Energy Consumption Test
1	L1 and L2	$V_{max} \leq 25 \text{ km/h}$	Part 1, RST25, followed by Part 1, RST25	2
		$25\text{km/h} < V_{max} \leq 50\text{km/h}$	Part 1 reduced speed followed by Part 1 reduced speed, if maximum design speed is $\leq 50 \text{ km/h}$ Part 1, RST45, followed by Part 1, RST45, if maximum design speed is $\leq 45 \text{ km/h}$	2
		$50 \text{ km/h} < V_{max} < 100 \text{ km/h}$	Part 1 reduced speed followed by Part 1 reduced speed	2

		$100 \text{ km/h} \leq V_{\max} < 115 \text{ km/h}$	Part 1 reduced speed followed by Part 2 reduced speed	2
		$115 \text{ km/h} \leq V_{\max} < 130 \text{ km/h}$	Part 1 followed by Part 2	2
		$130 \text{ km/h} \leq V_{\max} < 140 \text{ km/h}$	Part 1 followed by Part 2 followed by Part 3 reduced speed	2
		$V_{\max} \geq 140 \text{ km/h}$	Part 1 followed by Part 2 followed by Part 3	2

and

Application of 34 cycles of IDC (of 108 s duration each) as given in Annexure-II of CMVR for L5 and L7 category of vehicles

*Note: For RST 25 and RST45 driving cycles, refer Amendment 2 of AIS 049 Rev 1

2.0 Clause No 5.1.1

Substitute following text for existing text:

5.1.1 The driving cycle shall be World-harmonised Motorcycle Test Cycle (WMTC) stage 2 for L1 and L2 category vehicles as per AIS 137 Part 1 and the Indian Driving Cycle (IDC) as given in Annexure-II of CMVR for all L5 and L7 category vehicles.

*Note: For RST 25 and RST45 driving cycles, refer Amendment 2 of AIS 049 Rev 1

3.0 Clause No 5.2

Substitute following text for existing text:

5.2 Power setting of the chassis dynamometer

The procedure prescribed in the document MoRTH/TAP/CMVR115/116 or AIS 137 as applicable and as amended from time to time shall be adopted. Reference mass shall be taken as defined in AIS-049:2016 Rev 1 as amended and revised from time to time

4.0 Clause No 5.3.1

add following note at the end

Note : If vehicle has multiple driving modes and vehicle is not able to follow the specified test cycle speed as mentioned in Clause No 3.4 (b) in default mode then the test shall be carried out in closest mode wherein vehicle follows the specified WMTC test cycle speed.

5.0 Clause No 5.4.2

Substitute following text for existing text:

The chassis dynamometer shall be set as per the settings in point 5.2 above. Starting within 4 h from to , 02 cycles of World-harmonised Motorcycle Test Cycle (WMTC)* or 34 cycles of IDC (of 108 s duration each) or 22 cycles of Part-I of MIDC (of 195 s duration each) or 2 MIDC cycles each made of Part-I (four elementary urban cycles of 195 s duration each) and Part-II (an extra-urban cycle) or 25 cycle of Delhi Driving Cycle as applicable (refer para 5.1.1 or 5.1.2 above) are run. At the end, the covered distance (D) in km is recorded.

*If the manufacturer can demonstrate to the test agency that 02 cycles of World-harmonised Motorcycle Test Cycle (WMTC) test distance can physically not be attained by the vehicle, the test cycle shall be conducted once and subsequently followed by a partial second test run. The second test run may stop if the minimum state of charge of the REESS is reached as referred to in AIS 156.

For L1 and L2 category of vehicles, automatic head lamp or daytime running lamp shall be ON, during the test cycle.

For vehicles of other than L1 and L2 category, if DRL is installed on vehicle, DRL may be switched ON during test cycle as per vehicle manufacture's recommendation.

The vehicle tested shall be equipped with the daytime running lamp system that has the highest electrical energy consumption among the daytime running lamp systems, which are fitted by the manufacturer to vehicles.

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THE AUTOMOTIVE RESEARCH ASSOCIATION OF INDIA
P. B. NO. 832, PUNE 411 004

ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
GOVERNMENT OF INDIA

9th April 2026

Amendment No. 2 (04/2024)
To
AIS 039 (Rev 1): Electric Power Train Vehicles – Measurement of
Electrical Energy Consumption

1.0 Clause No 3.4 (b)

Substitute following text for existing text:

(b) Application of 34 cycles of IDC (of 108 s duration each) as given in Annexure-II of CMVR for L category of vehicles.

Or

2 Modified Indian Driving Cycles (MIDC) each made of Part-I (four elementary urban cycles of 195 s duration each) and Part-II (an extra-urban cycle) as given in Annexure-IV B of CMVR (test distance: 22 km, test duration: 40 minutes) for vehicles of category M1, M2 (with GVW up to 3500 kg).

Or

22 Cycles of Part-I of the Modified Indian Driving Cycle (MIDC) as given in Annexure-IV B of CMVR for N Category of vehicles.

Or

25 Cycles of Delhi Driving Cycle for M2 (with GVW above 3500 kg) and M3 category of vehicles (refer para 5.1.1 or 5.1.2).

2.0 Clause No 5.1.2

Substitute following text for existing text:

The driving cycle shall be Modified Indian Driving Cycles (MIDC) each made of Part-I (four elementary urban cycles of 195 s duration each) and Part-II (an extra-urban cycle) as given in Annexure-IV B of CMVR for vehicles of category M1, M2 (with GVW up to 3500 kg).

The driving cycle shall be, Part-I of the modified Indian Driving Cycle as given in Annexure-IV B of CMVR, for vehicles of category N

The driving cycle shall be Delhi Driving Cycle as given in AIS-049 (Rev 1):2016, as amended and revised from time to time, for M2 (with GVW above 3500 kg) and M3 category of vehicles.

3.0 Clause No 5.4.3

Add below text at the end of paragraph.

For vehicles with External DC Charging facility only (without onboard or portable charger supplied along with vehicle), the energy measurement equipment, shall be placed between External DC charger and vehicle

charging inlet socket to measures the DC charge energy E delivered from the charger to the vehicle.

4.0 Page No 5/8 Clause No 5.5.4 Test Results

Substitute following text for existing text:

The result of the electric energy consumption shall be expressed in Watt - hour per kilometer (Wh/km) rounded off to the nearest whole number.

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ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
GOVERNMENT OF INDIA

4th April 2024

Amendment 1 (11/2021)
to
AIS 039 (Rev 1) : Electric Power Train Vehicles– Measurement of Electrical Energy Consumption

1.0 Page No 2/8 Clause No 3.4 (b)

Substitute following text for existing text

(b) Application of 34 cycles of IDC (of 108 s duration each) as given in Annexure-II of CMVR for L category of vehicles

Or

For vehicles of category M1, M2 (with GVW up to 3500 kg), energy consumption shall be measured separately for “Urban” and “Urban+ Extra Urban” cycle by application of driving cycles,

- i. 22 Cycles of Part-I of the modified Indian driving cycle as given in Annexure-IV B of CMVR.

And

- ii. 2 modified Indian driving cycles (MIDC) each made of Part-I (four elementary urban cycles of 195 s duration each) and Part-II (an extra-urban cycle) as given in Annexure-IV B of CMVR (test distance: 22 km, test duration: 40 minutes)

Or

22 Cycles of Part-I of the modified Indian driving cycle as given in Annexure-IV B of CMVR for N Category of vehicles.

Or

25 Cycles of Delhi Driving Cycle for M2 (with GVW above 3500 kg) and M3 category of vehicles (refer para 5.1.1 or 5.1.2)

2.0 Page No 4/8 Clause No 5.1.2

Substitute following text for existing text:

For vehicles of category M1, M2 (with GVW up to 3500 kg), two separate energy consumption measurement tests shall be carried out as per driving cycles given below.

For vehicles of category M1, M2 (with GVW up to 3500 kg)	The driving cycle shall be
	i. Part-I of the modified Indian driving cycle as given in Annexure-IV B of CMVR.
	ii. Part-I (four elementary urban cycles of 195 s duration each) and Part-II (an extra-urban cycle) of the modified Indian driving cycle (MIDC) as given in Annexure-IV B of CMVR.

The driving cycle shall be, Part-I of the modified Indian driving cycle as given in Annexure-IV B of CMVR, for vehicles of category N

The driving cycle shall be Delhi Driving Cycle as given in AIS-049:2003, as amended and revised from time to time for M2 (with GVW above 3500 kg) and M3 category of vehicles.

3.0 Page No 4/8 Clause No 5.3.1

Substitute following text for existing text:

Principle

The test method described hereafter permits to measure the electric energy consumption expressed in Wh/km, to be measured:

If the Electric Power Train Vehicle has multiple driving modes,

where default mode is –

- a) available, test shall be carried out in default mode.
- b) not available, then test shall be carried out in worst case mode agreed between vehicle manufacture and test agency.

4.0 Page No 5/8 Clause No 5.4.2

Substitute following text for existing text:

The chassis dynamometer shall be set as per the settings in point 5.2 above. Starting within 4 h from to , 34 cycles of IDC (of 108 s duration each) or 22 cycles of Part-I of MIDC (of 195 s duration each) or 2 MIDC cycles each made of Part-I (four elementary urban cycles of 195 s duration each) and Part-II (an extra-urban cycle) or 25 cycle of Delhi Driving Cycle as applicable (refer para 5.1.1 or 5.1.2 above) are run. At the end, the covered distance (D) in km is recorded.

For L1 category of vehicles, automatic head lamp and for L2 category of vehicles, automatic head lamp or daytime running lamps shall be ON, during the test cycle.

For vehicles of other than L1 and L2 category, if DRL is installed on vehicle, DRL may be switched ON during test cycle as per vehicle manufacture's recommendation.

The vehicle tested shall be equipped with the daytime running lamp system that has the highest electrical energy consumption among the daytime running lamp systems, which are fitted by the manufacturer to vehicles.

5.0 Page No 5/8 Clause No 5.5.4

Add below text at the end of paragraph.

For vehicles of category M1, M2 (with GVW up to 3500 kg), result of the electric energy consumption expressed in Watt - hour per kilometer (Wh/km), measured with Part-I of the modified Indian driving cycle as given in Annexure-IV B of CMVR, is the electric energy consumption adopted as the type approved value of the electric vehicle.

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ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA

18th November 2021

CORRIGENDUM 1 5th September 2017
TO
AIS-039 (REV.1):2015: ELECTRIC POWER TRAIN VEHICLES–
MEASUREMENT OF ELECTRICAL ENERGY CONSUMPTION

1 Page No. 5/8, Clause 5.4.2

Substitute figure and words “25 cycles of Delhi Driving Cycle” for figure and words “1 cycles of Delhi Driving Cycle”.

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UNDER
CENTRAL MOTOR VEHICLES RULES - TECHNICAL STANDING COMMITTEE
SET-UP BY
MINISTRY OF ROAD TRANSPORT & HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT & HIGHWAYS)
GOVERNMENT OF INDIA
5th September 2017

AUTOMOTIVE INDUSTRY STANDARDS

**Electric Power Train Vehicles—
Measurement of Electrical Energy
Consumption**

(Revision 1)

PRINTED BY
THE AUTOMOTIVE RESEARCH ASSOCIATION OF INDIA
P.B. NO. 832, PUNE 411 004

ON BEHALF OF
AUTOMOTIVE INDUSTRY STANDARDS COMMITTEE

UNDER
CENTRAL MOTOR VEHICLE RULES – TECHNICAL STANDING COMMITTEE

SET-UP BY
MINISTRY OF ROAD TRANSPORT AND HIGHWAYS
(DEPARTMENT OF ROAD TRANSPORT AND HIGHWAYS)
GOVERNMENT OF INDIA

February 2015

**Status chart of the Standard to be used by the Purchaser for
updating the record.**

Sr.	Corrigenda	Amend- ment	Revision	Date	Remark	Misc.

General Remarks:

INTRODUCTION

The Government of India felt the need for a permanent agency to expedite the publication of Standards and development of test facilities in parallel when the work of preparation of Standards is going on, as the development of improved safety critical parts can be undertaken only after the publication of the Standard and commissioning of test facilities. To this end, the erstwhile Ministry of Surface Transport (MOST) has constituted a permanent Automotive Industry Standard Committee (AISC) vide order no. RT-11028/11/97-MVL dated September 15, 1997. The Standards prepared by AISC will be approved by the permanent CMVR Technical Standing Committee (CTSC) after approval, The Automotive Research Association of India, (ARAI), Pune, being the secretariat of the AIS Committee, has published this Standard. For better dissemination of this information, ARAI may publish this standard on their website.

This standard prescribes the requirements for the measurement of electrical energy consumption.

Considerable assistance has been taken from UN R101 Rev 3 (Supplement 1 to the 01 series of amendments - Date of entry into force: 27 January 2013): Uniform Provisions concerning the approval of Passenger Cars powered by an Internal Combustion Engine only, or powered by a Hybrid Electric Power Train with regard to the measurement of the Emission of Carbon Dioxide and Fuel Consumption and/or the measurement of Electric Energy Consumption and Electric Range, and of categories M1 and N1 vehicles powered by an Electric Power Train only with regard to the measurement of Electric Energy Consumption and Electric Range.

The AISC panel and the Automotive Industry Standards Committee (AISC) responsible for preparation of this standard are given in Annex A and Annex B respectively.

Electric Power Train Vehicles - Measurement of Electrical Energy Consumption

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Electric Power Train Vehicles - Measurement of Electrical Energy Consumption

1.0 SCOPE

This standard specifies the method for measurement of electrical energy consumption expressed in Wh/km for L, M & N categories of Electric Power Train Vehicles as defined in Rule 2 (u) of CMVR.

2.0 DEFINITIONS

Refer AIS-049:2003, as amended and revised from time to time, for the definitions.

3.0 VEHICLE PREPARATION

3.1 Condition of the Vehicle

The vehicle tyres shall be inflated to the pressure specified by the vehicle manufacturer when tyres are at the ambient temperature. The lubricants for the mechanical moving parts shall conform to the specifications of the vehicle manufacturer. The lighting, light – signaling and auxiliary devices shall be off, except those required for testing and usual daytime operation of the vehicle.

3.2 All energy storage systems available for other than traction purpose (electric, hydraulic, pneumatic, etc.) shall be charged up to their maximum level specified by the manufacturer.

3.3 If the Rechargeable Energy Storage System (REESS) is operated above the ambient temperature, the operator shall follow the procedure recommended by the vehicle manufacturer in order to keep the temperature of the REESS in the normal operating range.

The manufacturer's representative shall be in a position to attest that the thermal management system of the REESS is neither disabled nor reduced.

The vehicle must have undergone at least 300 km during the seven days before the test with those REESS that are installed in the test vehicle. The seven days period can be waived on the request of vehicle manufacturer.

3.4 Operation mode

All the tests are conducted at a temperature of between 20 °C and 30 °C.

Note: If the ambient condition cannot be met at the time of the test, then based on mutual agreement between test agency and vehicle manufacturer, requirement of ambient temperature condition can be waived.

The test method includes the four following steps:

- (a) Initial charge of the Rechargeable Energy Storage System (REESS);
- (b) Application of 34 cycles of IDC (of 108 s duration each) for
L category of vehicles

Or

22 Cycles of Part 1 of MIDC (of 195 s duration each) for M1, M2 (with GVW upto 3500 kg) and N category of vehicles.

Or

25 Cycles of Delhi Driving Cycle for M2 (with GVW above 3500 kg) and M3 category of vehicles (refer para 5.1.1 or 5.1.2)

- (c) Charging the REESS;
- (d) Calculation of the electric energy consumption.

Between the steps, if the vehicle needs to be moved, it is pushed to the following test area (without regenerative recharging).

3.5 Initial charge of the Rechargeable Energy Storage System (REESS)

Charging the REESS consists of the following procedures.

Note: “Initial charge of the REESS” applies to the first charge of the REESS, at the reception of the vehicle. In case of several combined tests or measurements, carried out consecutively, the first charge carried out shall be an “initial charge of the REESS” and the following may be done in accordance with the “normal overnight charge” procedure.

3.5.1 Discharge of the Rechargeable Energy Storage System (REESS)

The procedure starts with the discharge of the battery of the vehicle while driving (on the test track, on a chassis dynamometer etc.) at a steady speed of $70\% \pm 5\%$ of the maximum thirty minutes speed of the vehicle. Stopping the discharge occurs:

- a) When the vehicle is not able to run at 65 % of the maximum thirty minutes speed or
- b) When an indication to stop the vehicle is given to the driver by the standard on-board instrumentation or
- c) After covering the distance of 100 km.

3.5.2 **Application of a normal overnight charge**

The Rechargeable Energy Storage System (REESS) shall be charged according to the following procedure.

3.5.2.1 Normal overnight charge procedure

The charge is carried out

- a) with the on-board charger, if fitted,
- b) with an external charger recommended by the manufacturer using the charging pattern prescribed for normal charging,
- c) in an ambient temperature comprised between 20 °C and 30 °C.

The procedure excludes all type of special charges that could be automatically or manually initiated like, for instance, the equalization charges or the servicing charges. The vehicle manufacturer shall declare that during the test, a special charge procedure has not occurred.

Note: If the ambient condition cannot be met at the time of the test, then based on mutual agreement between test agency and vehicle manufacturer requirement of ambient temperature condition can be waived.

3.5.2.2 **End of charge criteria**

The end of charge criteria corresponds to a charging time of 12 h except if a clear indication is given to the driver by the standard instrumentation that the REESS is not yet fully charged or as specified by the manufacturer. In this case

The maximum time is =

$$\frac{3 \times \text{Claimed battery capacity (Wh)}}{\text{Mains power supply (W)}}$$

4.0 **FULLY CHARGED RECHARGEABLE ENERGY STORAGE SYSTEM (REESS)**

REESS having been charged according to overnight charge procedure until the end of charge criteria.

5.0 **TEST PROCEDURE**

5.1 **Test Sequence**

5.1.1 The driving cycle shall be the Indian Driving Cycle (IDC) as given in Annexure-II of CMVR for all L category vehicles.

5.1.2 The driving cycle shall be Part-I of the modified Indian driving cycle as given in Annexure-IV B of CMVR for M1, M2 (with GVW upto 3500 kg) and N category of vehicles.

The driving cycle shall be Delhi Driving Cycle as given in AIS-049:2003, as amended and revised from time to time for M2 (with GVW above 3500 kg) and M3 category of vehicles.

5.1.3 In cases where the vehicle does not reach the required acceleration during driving, the accelerator/speed control shall remain fully activated until the reference curve has been reached again.

5.2 Power setting of the chassis dynamometer

The procedure prescribed in the document MoRTH/TAP/CMVR-115/116 as amended from time to time shall be adopted. Reference mass shall be taken as defined in AIS-049:2003 as amended and revised from time to time

5.3 Test Method

5.3.1 Principle

The test method described hereafter permits to measure the electric energy consumption expressed in Wh/km, to be measured:

5.3.2 Parameters, Units and Accuracy of Measurements

Parameter	Unit	Accuracy	Resolution
Time	s	+/-0.1 s	0.1 s
Distance	m	+/-0.1 per cent	1 m
Temperature	°C	+/-1 degree C	1 degree C
Speed	km/h	+/-1 per cent	0.2 km/h
Mass	kg	+/-0.5 per cent	1 kg
Energy	Wh	+/-0.5 per cent	0.3 per cent
IEC: International Electrotechnical Commission.			
Where accuracy is specified in %, it is the % of the measured value.			

5.4 Application of the Cycle and Measurement of the Distance

5.4.1 The end of charging t_0 (plug off) is reported.

5.4.2 The chassis dynamometer shall be set as per the settings in point 5.2 above. Starting within 4 h from t_0 , 34 cycles of IDC (of 108 s duration each) or 22 cycles of Part1 of MIDC (of 195 s duration each) or 1 cycle of Delhi Driving Cycle as applicable (refer para 5.1.1 or 5.1.2 above) are run. At the end, the covered distance (D) in km is recorded.

5.4.3 **Charge of the Rechargeable Energy Storage System (REESS):**

The vehicle shall be connected to the mains within 30 minutes after the conclusion of the Driving Cycle. The vehicle shall be charged according to normal overnight charge procedure (Refer clause 3.5.2.1 above). The energy measurement equipment, placed between the mains socket and the vehicle charger, measures the charge energy E delivered from the mains as well as its duration. Charging is stopped after 24 h from the previous end of charging time t_0 .

Note: In case of any power disruptions during charging, the 24 h period shall be exceeded according to the disruption duration. The maximum total Power disruption of 30 minutes duration is allowed irrespective of the number of failures. Validity of the charge shall be discussed between the Test Agency and the vehicle manufacturer.

5.4.4 **Electric Energy Consumption Calculation**

Energy E in Wh and charging time measurements shall be recorded in the test report. The electric energy consumption is defined by the formula :

$$C = E / D$$

Where C is expressed in Wh/km and rounded off to the nearest whole number, E is Wh and D is the distance covered during the test in km.

5.5 **Interpretation of the Results**

5.5.1 The electric energy consumption adopted as the type approved value, shall be the value declared by the manufacturer if the value measured during testing does not exceed the declared value by more than 4%. The measured value can be lower without any limitations.

5.5.2 If the measured value of energy consumption exceeds the manufacturers declared value by more than 4% then another test shall be carried out on the same vehicle. When the average of the two test results does not exceed the manufacturer's declared value, by more than 4 % then the value declared by the manufacturer shall be taken as the type approval value.

5.5.3 If the average still exceeds the declared value by more than 4%, a final test shall be run on the same vehicle. The average of the three tests shall be then taken as the type approval value.

5.5.4 **Test Results**

The result of the electric energy consumption shall be expressed in Watt - hour per kilometer (Wh/km) rounded off to the nearest whole

number.

6.0 TECHNICAL SPECIFICATIONS

The details of technical specification, approvals of changes in specification shall be as per para 5.0 of AIS-049:2003 as amended and revised from time to time.

7.0 TRANSITIONAL PROVISIONS

7.1 General guidelines for transitional provisions for this standard shall be as per AIS-000, as amended from time to time, as applicable, with the following requirement.

7.2 For L category vehicles type approvals issued for compliance to AIS-039:2003 shall be extended for approval to of AIS-039 (Rev1):2015. A retest may be performed at the request of manufacturer.

7.3 For M and N category vehicles re-testing shall be carried out for compliance to AIS-039 (Rev 1): 2015 from the date of notification. Manufacturers may request testing as per this standard from the date of adoption of this standard in CMVR-TSC.

ANNEX A
(See Introduction)

**COMPOSITION OF AISC PANEL ON
ELECTRIC POWER TRAIN VEHICLES– MEASUREMENT OF
ELECTRICAL ENERGY CONSUMPTION**

Convener	
Mr. A.A. Deshpande	The Automotive Research Association of India (ARAI)
Members	Representing
Mr. M. M. Desai	The Automotive Research Association of India (ARAI)
Mr. D. P. Saste/ Mr. Karthikeyan K (Alternate)	Central Institute of Road Transport (CIRT)
Representative from	International Centre for Automotive Technology (ICAT)
Mr. Vinod Kumar	Vehicle Research & Dev. Estt. (VRDE)
Dr. N. Karuppaiah	National Automotive Testing and R&D Infrastructure Project (NATRIP)
Mr. K. K. Gandhi	Society of Indian Automobile Manufacturers (SIAM)
Mr. T. M. Balaraman	Society of Indian Automobile Manufacturers (SIAM) (Hero MotoCorp Ltd.)
Mr. Adish Agrawal	Society of Indian Automobile Manufacturers (SIAM) (Hero MotoCorp Ltd.)
Mr. Rajendra Khile	Society of Indian Automobile Manufacturers (SIAM) (General Motors)
Mr. Kiran Mulki	Society of Indian Automobile Manufacturers (SIAM) (Mahindra and Mahindra)
Mr. K. Kiran Kumar	Society of Indian Automobile Manufacturers (SIAM) (Mahindra Reva)
Mr. Vijeth R Gatty	Society of Indian Automobile Manufacturers (SIAM) (Toyota Kirloskar Motors Ltd.)
Mr. Firoz Khan	Society of Indian Automobile Manufacturers (SIAM) (Tata Motors Ltd.)
Mr. Vivekraj S	Society of Indian Automobile Manufacturers (SIAM) (Renault Nissan)
Mr. P. C. Joshi	Bureau of Indian Standards (BIS)

* At the time of approval of this Automotive Industry Standard (AIS)

ANNEX B
(See Introduction)
COMMITTEE COMPOSITION *
Automotive Industry Standards Committee

Chairperson	
Mrs. Rashmi Urdhwareshe	Director The Automotive Research Association of India, Pune
Members	Representing
Representative from	Ministry of Road Transport and Highways (Dept. of Road Transport and Highways), New Delhi
Representative from	Ministry of Heavy Industries and Public Enterprises (Department of Heavy Industry), New Delhi
Shri S. M. Ahuja	Office of the Development Commissioner, MSME, Ministry of Micro, Small and Medium Enterprises, New Delhi
Shri Shrikant R. Marathe	Former Chairman, AISC
Shri N. K. Sharma	Bureau of Indian Standards, New Delhi
Director/ Shri D. P. Saste (Alternate)	Central Institute of Road Transport, Pune
Director	Indian Institute of Petroleum, Dehra Dun
Director	Vehicles Research and Development Establishment, Ahmednagar
Representatives from	Society of Indian Automobile Manufacturers
Shri T. C. Gopalan	Tractor Manufacturers Association, Chennai
Shri Uday Harite	Automotive Components Manufacturers Association of India, New Delhi

Member Secretary
Mr. A. S. Bhale
General Manager
The Automotive Research Association of India, Pune

* At the time of approval of this Automotive Industry Standard (AIS)